

Cloud Computing

ASSIGNMENT 1

Submitted To:

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SoC - Department of Software Engineering

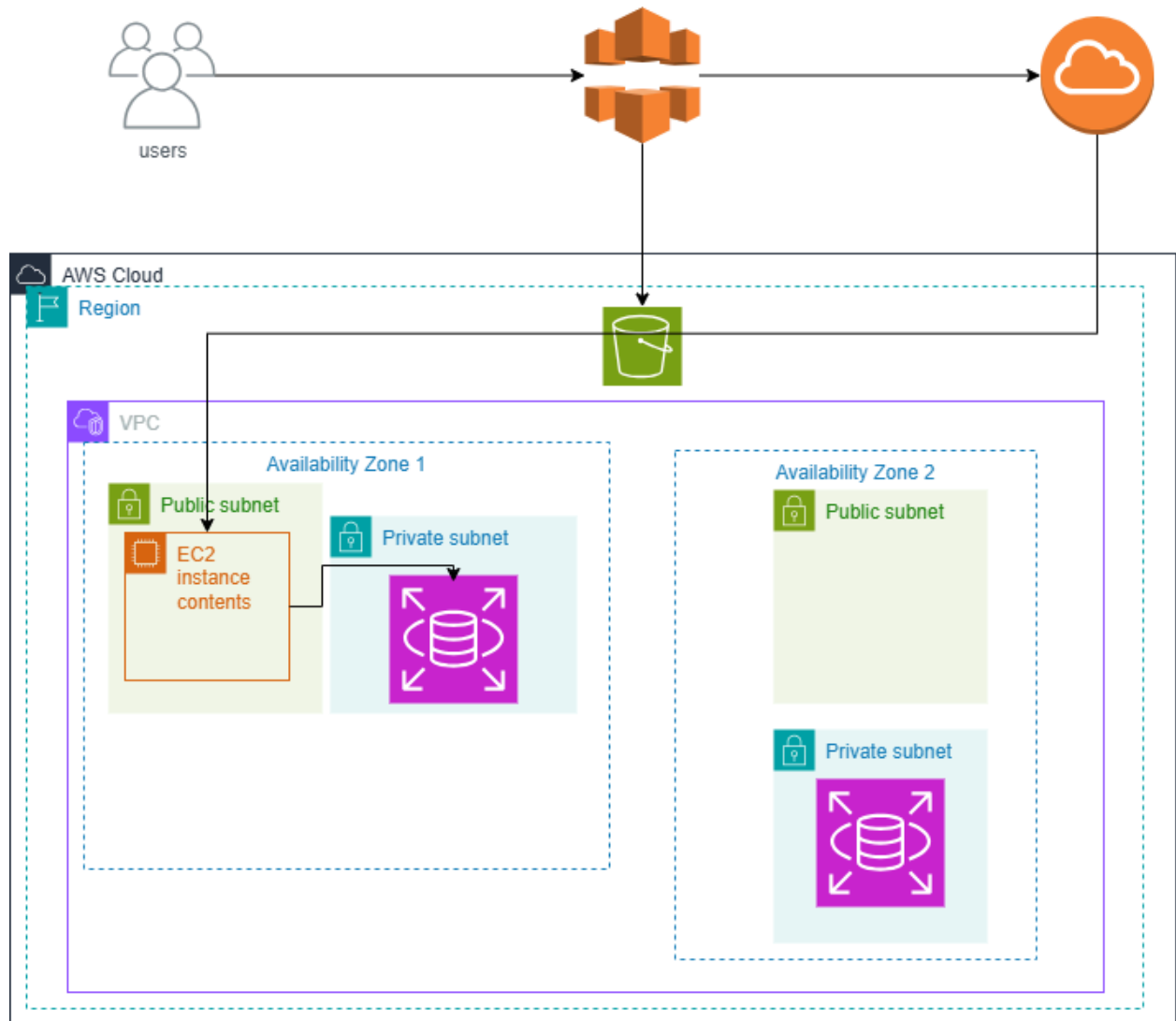
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1. Deployment Architecture:



2. Screenshot and Analysis of Region's Latency from Pakistan

Region	Latency
<u>Amazon Web Services™</u>	
us-east-1 (Virginia)	391 ms
us-east-2 (Ohio)	397 ms
us-west-1 (California)	773 ms
us-west-2 (Oregon)	773 ms
ca-central-1 (Canada Central)	310 ms
ca-west-1 (Canada West)	881 ms
eu-west-1 (Ireland)	939 ms
eu-west-2 (London)	940 ms
eu-west-3 (Paris)	timeout
eu-central-1 (Frankfurt)	timeout
eu-central-2 (Zurich)	406 ms
eu-south-1 (Milan)	172 ms
eu-south-2 (Spain)	193 ms
eu-north-1 (Stockholm)	197 ms

il-central-1 (Doing business with Israel supports atrocities.)	202 ms
me-south-1 (Doing business with Bahrain supports slavery.)	69 ms
me-central-1 (Doing business with the UAE supports slavery.)	199 ms
af-south-1 (Cape Town)	427 ms
ap-east-1 (Hong Kong)	538 ms
ap-east-2 (Taipei)	544 ms
ap-southeast-3 (Jakarta)	1402 ms
ap-south-1 (Mumbai)	153 ms
ap-south-2 (Hyderabad)	322 ms
ap-northeast-3 (Osaka)	305 ms
ap-northeast-2 (Seoul)	293 ms
ap-southeast-1 (Singapore)	171 ms
ap-southeast-2 (Sydney)	240 ms
ap-southeast-4 (Melbourne)	245 ms
ap-northeast-1 (Tokyo)	226 ms
sa-east-1 (São Paulo)	467 ms

I used [CloudPing.info](#) to check the latency across different servers and Analyzed that the Middle East (Bahrain) us-east-1 has a minimum time of 69ms.

Summary of the selected services and their configuration:

Region North America (US East (N. Virginia))

Amazon EC2:

Tenancy (Shared Instances)

Operating system (Linux)

Workload (Consistent, Number of instances: 1)

Advance EC2 instance (t3.medium)

Pricing strategy (Compute Savings Plans 3yr No Upfront)

Enable monitoring (disabled), DT Inbound: Internet (0 TB per month)

DT Outbound: All other regions (100 GB per month)

DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)

Storage volume (General Purpose SSD (gp2))

Nodes (1)

Instance Type (db.t3.medium)

Utilization (On-Demand only) (100 %Utilized/Month)

Deployment Option (Multi-AZ)

Pricing Model (OnDemand)

Standard EC2:

S3 Standard storage (500 GB per month)

PUT, COPY, POST, LIST requests to S3 Standard (10000)

GET, SELECT, and all other requests from S3 Standard (100000)

Amazon CloudFront:

Business Plan (40)

Upfront cost

0.00 USD

Monthly cost

8,197.63 USD

Total 12 months cost

98,371.56 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	17.04 USD	-	US East (N. Virginia)	Tenancy (Shared Instances), Operating sy...
☐	Amazon RDS for ...	-	0.00 USD	169.00 USD	-	US East (N. Virginia)	Storage amount (100 GB), Storage volum...
☐	Amazon Simple S...	-	0.00 USD	11.59 USD	-	US East (N. Virginia)	S3 Standard storage (500 GB per month),...
☐	Amazon CloudFront	-	0.00 USD	8,000.00 USD	-	US East (N. Virginia)	Free Plan (), Pro Plan (), Business Plan (40)

Amazon EC2:

Tenancy (Shared Instances)

Operating system (Linux)

Workload (Consistent, Number of instances: 1)

Advance EC2 instance (t3.medium)

Pricing strategy (On-Demand Utilization: 100 %Utilized/Month)

Enable monitoring (disabled)

DT Inbound: Internet (0 TB per month)

DT Outbound: All other regions (100 GB per month)

DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)

Storage volume (General Purpose SSD (gp2))

Nodes (1), Instance Type (db.t3.medium)

Utilization (On-Demand only) (100 %Utilized/Month)

Deployment Option (Multi-AZ), Pricing Model (Reserved)

Lease Offering Term (1yr)

Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

Selected IA

Amazon CloudFront:

Free Plan (3), Pro Plan (100), Business Plan ()

Upfront cost

0.00 USD

Monthly cost

1,671.00 USD

Total 12 months cost

20,052.00 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	32.37 USD	-	US East (N. Virginia)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	138.63 USD	-	US East (N. Virginia)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	US East (N. Virginia)	
☐	Amazon CloudFront	-	0.00 USD	1,500.00 USD	-	US East (N. Virginia)	Free Plan (3), Pro Pla...

Region Europe (London)

Amazon EC2:

Tenancy (Shared Instances)

Operating system (Linux)

Workload (Consistent, Number of instances: 1)

Advance EC2 instance (t3.medium)

Pricing strategy (On-Demand Utilization: 100 %Utilized/Month)

Enable monitoring (disabled)

DT Inbound: Internet (0 TB per month)

DT Outbound: All other regions (100 GB per month)

DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)

Storage volume (General Purpose SSD (gp2))

Nodes (1), Instance Type (db.t3.medium)

Utilization (On-Demand only) (100 %Utilized/Month)

Deployment Option (Multi-AZ), Pricing Model (Reserved)

Lease Offering Term (1yr)

Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

Selected IA

Amazon CloudFront:

Free Plan (3), Pro Plan (100), Business Plan ()

Upfront cost

0.00 USD

Monthly cost

1,694.10 USD

Total 12 months cost

20,329.20 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	36.46 USD	-	Europe (London)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	157.64 USD	-	Europe (London)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	Europe (London)	
☐	Amazon CloudFront	-	0.00 USD	1,500.00 USD	-	Europe (London)	Free Plan (3), Pro Pla...

Amazon EC2:

Tenancy (Shared Instances)

Operating system (Linux)

Workload (Consistent, Number of instances: 1)

Advance EC2 instance (t3.medium)

Pricing strategy (Compute Savings Plans 3yr No Upfront)

Enable monitoring (disabled), DT Inbound: Internet (0 TB per month)

DT Outbound: All other regions (100 GB per month)

DT Intra-Region: (0 TB per month)

DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)

Storage volume (General Purpose SSD (gp2))

Nodes (1), Instance Type (db.t3.medium)

Utilization (On-Demand only) (100 %Utilized/Month)

Deployment Option (Multi-AZ), Pricing Model (Reserved)

Lease Offering Term (1yr)
Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

S3 Glacier Flexible Retrieval Average Object Size (16 MB)

Amazon CloudFront:

Free Plan (3), Pro Plan (100), Business Plan ()

Upfront cost

0.00 USD

Monthly cost

1,681.53 USD

Total 12 months cost

20,178.36 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	23.89 USD	-	Europe (London)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	157.64 USD	-	Europe (London)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	Europe (London)	S3 Glacier Flexible Re...
☐	Amazon CloudFront	-	0.00 USD	1,500.00 USD	-	Europe (London)	Free Plan (3), Pro Plan (100)

Region Middle East (UAE)

Amazon EC2:

Tenancy (Shared Instances)
Operating system (Linux)
Workload (Consistent, Number of instances: 1)
Advance EC2 instance (t3.medium)
Pricing strategy (On-Demand Utilization: 100 %Utilized/Month)
Enable monitoring (disabled)
DT Inbound: Internet (0 TB per month)
DT Outbound: All other regions (100 GB per month)
DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)

Storage volume (General Purpose SSD (gp2))
Nodes (1), Instance Type (db.t3.medium)
Utilization (On-Demand only) (100 %Utilized/Month)
Deployment Option (Multi-AZ), Pricing Model (Reserved)
Lease Offering Term (1yr)
Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

Selected IA

Amazon CloudFront:

We do not have CloudFront in this region

Upfront cost

0.00 USD

Monthly cost

207.88 USD

Total 12 months cost

2,494.56 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	45.15 USD	-	Middle East (UAE)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	162.73 USD	-	Middle East (UAE)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	Middle East (UAE)	

Amazon EC2:

Tenancy (Shared Instances)
Operating system (Linux)
Workload (Consistent, Number of instances: 1)
Advance EC2 instance (t3.medium)
Pricing strategy (Compute Savings Plans 3yr No Upfront)
Enable monitoring (disabled), DT Inbound: Internet (0 TB per month)
DT Outbound: All other regions (100 GB per month)
DT Intra-Region: (0 TB per month)
DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)
Storage volume (General Purpose SSD (gp2))
Nodes (1), Instance Type (db.t3.medium)
Utilization (On-Demand only) (100 %Utilized/Month)
Deployment Option (Multi-AZ), Pricing Model (Reserved)
Lease Offering Term (1yr)
Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

S3 Glacier Flexible Retrieval Average Object Size (16 MB)

Amazon CloudFront:

Does not have a cloudfront option

Upfront cost

0.00 USD

Monthly cost

181.53 USD

Total 12 months cost

2,178.36 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	23.89 USD	-	Europe (London)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	157.64 USD	-	Europe (London)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	Europe (London)	S3 Glacier Flexible Re...

Region Asia Pacific (Singapore)

Amazon EC2:

Tenancy (Shared Instances)
Operating system (Linux)
Workload (Consistent, Number of instances: 1)
Advance EC2 instance (t3.medium)
Pricing strategy (On-Demand Utilization: 100 %Utilized/Month)
Enable monitoring (disabled)
DT Inbound: Internet (0 TB per month)
DT Outbound: All other regions (100 GB per month)

DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)

Storage volume (General Purpose SSD (gp2))

Nodes (1), Instance Type (db.t3.medium)

Utilization (On-Demand only) (100 %Utilized/Month)

Deployment Option (Multi-AZ), Pricing Model (Reserved)

Lease Offering Term (1yr)

Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

Selected IA

Amazon CloudFront:

Free Plan (3), Pro Plan (100), Business Plan ()

Upfront cost

0.00 USD

Monthly cost

1,746.55 USD

Total 12 months cost

20,958.60 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	45.65 USD	-	Asia Pacific (Singapore)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	200.90 USD	-	Asia Pacific (Singapore)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	Asia Pacific (Singapore)	
☐	Amazon CloudFront	-	0.00 USD	1,500.00 USD	-	Asia Pacific (Singapore)	Free Plan (3), Pro Plan (100)

Amazon EC2:

Tenancy (Shared Instances)

Operating system (Linux)

Workload (Consistent, Number of instances: 1)

Advance EC2 instance (t3.medium)

Pricing strategy (On-Demand Utilization: 100 %Utilized/Month)

Enable monitoring (disabled)

DT Inbound: Internet (0 TB per month)
DT Outbound: All other regions (100 GB per month)
DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)
Storage volume (General Purpose SSD (gp2))
Nodes (1), Instance Type (db.t3.medium)
Utilization (On-Demand only) (100 %Utilized/Month)
Deployment Option (Multi-AZ), Pricing Model (Reserved)
Lease Offering Term (1yr)
Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

S3 Glacier Flexible Retrieval Average Object Size (16 MB)

Amazon CloudFront:

Free Plan (3), Pro Plan (100), Business Plan ()

Upfront cost

0.00 USD

Monthly cost

1,748.83 USD

Total 12 months cost

20,985.96 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	47.93 USD	-	Asia Pacific (Singapore)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	200.90 USD	-	Asia Pacific (Singapore)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	Asia Pacific (Singapore)	S3 Glacier Flexible Re...
☐	Amazon CloudFront	-	0.00 USD	1,500.00 USD	-	Asia Pacific (Singapore)	Free Plan (3), Pro Plan (100)

Amazon EC2:

Tenancy (Shared Instances)
Operating system (Linux)
Workload (Consistent, Number of instances: 1)
Advance EC2 instance (t3.medium)
Pricing strategy (Compute Savings Plans 3yr No Upfront)

Enable monitoring (disabled), DT Inbound: Internet (0 TB per month)
DT Outbound: All other regions (100 GB per month)
DT Intra-Region: (0 TB per month)
DT Intra-Region: (0 TB per month)

Amazon RDS:

Storage amount (100 GB)
Storage volume (General Purpose SSD (gp2))
Nodes (1), Instance Type (db.t3.medium)
Utilization (On-Demand only) (100 %Utilized/Month)
Deployment Option (Multi-AZ), Pricing Model (Reserved)
Lease Offering Term (1yr)
Purchase Option (No Upfront)

Amazon Simple Storage Service (S3)

S3 Glacier Flexible Retrieval Average Object Size (16 MB)

Amazon CloudFront:

Free Plan (3), Pro Plan (100), Business Plan ()

Upfront cost

0.00 USD

Monthly cost

1,726.25 USD

Total 12 months cost

20,715.00 USD

Includes upfront cost

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☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	25.35 USD	-	Asia Pacific (Singapore)	Tenancy (Shared Inst...
☐	Amazon RDS for Post...	-	0.00 USD	200.90 USD	-	Asia Pacific (Singapore)	Storage amount (10...
☐	Amazon Simple Stor...	-	0.00 USD	0.00 USD	-	Asia Pacific (Singapore)	S3 Glacier Flexible Re...
☐	Amazon CloudFront	-	0.00 USD	1,500.00 USD	-	Asia Pacific (Singapore)	Free Plan (3), Pro Plan (100)

Critical Details:

CloudFront costs are global, but for the purpose of this regional comparison, the UAE estimate reflects only the core infrastructure cost differences.

Comparison Table:

Region	EC2 (Compute)	RDS (Database)	Data Transfer & Misc.	Total Monthly Cost
US East (N. Virginia)	\$32.40	\$138.63	\$95.00	\$266.03
Europe (London)	\$41.75	\$165.10	\$102.50	\$309.35
Asia Pacific (Singapore)	\$45.30	\$200.90	\$110.00	\$356.20
Middle East (UAE)	\$45.15	\$182.40	\$108.00	\$335.55

Cost Optimization Strategy

Optimization Method	Monthly Cost	Savings	Notes
On-Demand Pricing	\$1,694.10	—	Baseline cost
Compute Savings Plan (1-Year, No Upfront)	\$1,681.53	\$12.57	Lower hourly compute rates
Use S3 Glacier for Archival	Further reduction	Variable	Store infrequently accessed images
Move background jobs to Spot Instances	Up to 90% compute savings	Significant	Ideal for non-critical workloads

Key Insights

- **Lowest Cost:** US East (N. Virginia) remains the most economical region.
- **Highest Cost:** Singapore shows the highest pricing, mainly due to higher RDS costs.
- **Latency vs Cost Tradeoff:** UAE provides better latency for Pakistan users but at a higher cost.
- **Optimization Impact:** Savings Plans and storage tiering provide predictable savings, while Spot Instances offer major reductions for flexible workloads.